A Bayesian Standard Model of Early Word Learning

Joint inference across vocabulary, input, and processing data

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April 2026

Goal: model how language input drives vocabulary learning

A unified statistical model of early word learning, organized around four research questions:

RQ1. How well does word frequency reconstruct word difficulty?
i.e., is $\psi_j \approx -\log p_j$ *enough?*

RQ2. When does word learning start?
is start time s *identifiable from data?*

RQ3. Does vocabulary growth accelerate over development?
is δ *greater than 0?*

RQ4. How much of between-child vocabulary variance is driven by input rate vs. learning efficiency?
what is $\pi_\alpha = \sigma_\alpha^2/(\sigma_\alpha^2 + \sigma_r^2)$*?*

Follow-up to Kachergis, Marchman, & Frank (2021): "Toward a standard model of early language learning."

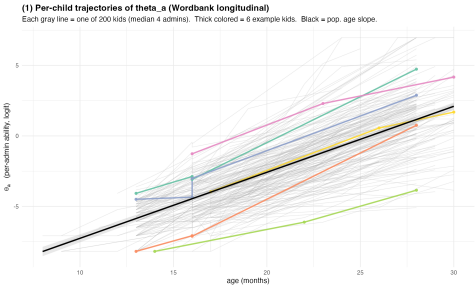
Motivation: how much variation is there to explain?

Fit the simplest possible 1-PL IRT — $P(y_{aj} = 1) = \sigma(\theta_a - \beta_j)$ — to the longitudinal Wordbank data. No accumulator dynamics, just per-admin θ_a, per-item β_j.

Population age slope: 0.47 logits/mo.
Within-age SD (θ_a): 1.95 logits.

Translation:

$$\underbrace{\frac{\mathrm{SD}_{\mathrm{within-age}}}{\hat{\beta}_{\mathrm{age}}}}_{\text{months / SD}} = \frac{1.95}{0.47} \approx \textbf{4.2 months}$$



Per-admin θ_a for each of ∼200 English children. Median 4 admins/kid. Black: pop. age slope.

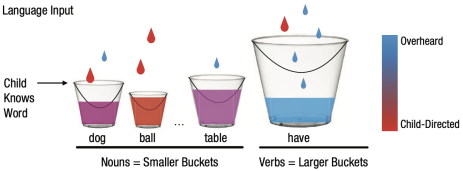
And it grows with age: 1.0 logit at 12–14 mo (∼2 mo) → 2.7 logits at 26–30 mo (∼5.8 mo). Children fan out, not converge.

Why this matters for the model: A pure accumulator with constant per-child rate cannot produce

Schematic model: bucket-fill metaphor + item-response theory

Two ideas from Kachergis et al. (2021):

1. *Accumulator.* Each word is a bucket; child hears tokens at rate $r_i \cdot p_j$; efficiency α_i determines fraction retained.
2. *Item-Response Theory.* Per-word difficulty ψ_j controls how many "filled" tokens it takes for the child to know the word.



Model: linear predictor (anatomy)

$P(\text{produces}_{ijt} = 1) = \text{logit}^{-1}(\eta_{ijt})$ where:

$$\eta_{ijt} = \underbrace{\lambda_j}_{\substack{\textit{item}\\ \textit{discrimination}\\ \textit{(2PL)}}} \left[\underbrace{\xi_i}_{\substack{\textit{child intercept}\\ = \log r_i + \log \alpha_i}} + \underbrace{\log p_j}_{\substack{\textit{word}\\ \textit{frequency}}} + \underbrace{\log H}_{\substack{\textit{waking}\\ \textit{hours/mo}}} + \underbrace{(1+\delta+\zeta_i)}_{\substack{\textit{1 + pop. accel.}\\ \textit{δ + per-child}\\ \textit{slope dev. ζ_i}}} \cdot \underbrace{\log \frac{t-s}{a_0}}_{\substack{\textit{log relative age}\\ \textit{(start time s)}}} - \underbrace{\psi_j}_{\substack{\textit{word}\\ \textit{difficulty}}} \right]$$

Factorization: ability vs. difficulty

The same equation regrouped as a 2PL IRT factorization:

$$\eta_{ijt} \;=\; \lambda_j \cdot \big(\, \theta_{it} \;-\; \beta_j \,\big)$$

Ability (child + age):

$$\theta_{it} = \xi_i + (1{+}\delta{+}\zeta_i)\,\log\tfrac{t-s}{a_0}$$

pop. accel. δ
+ per-child ζ_i

$$\xi_i \;+\; (1{+}\delta{+}\zeta_i) \;\cdot\; \log\tfrac{t-s}{a_0}$$

child intercept
$\log r_i + \log \alpha_i$

start time s

Difficulty (item):

$$\beta_j = \psi_j - \log p_j - \log H$$

per-word log
threshold tokens
(class-hierarchical)

$$\psi_j \;-\; \log p_j \;-\; \log H$$

corpus frequency
(unit coefficient)

hours/month
(constant)

Why split it. $\log p_j$ and $\log H$ together convert "probability" into "expected tokens per month at unit

Model variants

Per-child latents. $(\log \alpha_i, \zeta_i) \sim \mathrm{MVN}(0, \Sigma)$ with LKJ prior on the correlation; $\log r_i \sim N(\mu_r, \sigma_r)$ independent. *Sum-to-zero centering on each random effect* identifies δ (population slope) separately from $\mathrm{mean}(\zeta_i)$.

Lean baseline + opt-in variants. Pin $s = 0$, $\sigma_\lambda = 0$, $\sigma_\zeta = 0$ by default; opt in explicitly:

variant	adds
`slopes`	ζ_i, σ_ζ (longitudinal default)
`2pl`	λ_j, σ_λ
`free_s`	free s
`fix_delta`	$\delta = 0$

σ_r **pinned externally** from Sperry / Hart-Risley / Weisleder-Fernald data; lets the model decompose $\sigma_\xi^2 = \sigma_r^2 + \sigma_\alpha^2$ and report π_α.

Main longitudinal sample: Wordbank English + Norwegian

Data. Wordbank longitudinal CDIs (WG $\cup$ WS forms). Children with ≥ 2 admins; both forms count toward the same child.

	N children	N admins	age range (mo)
English (American)	200	$\sim$808	8–32
Norwegian	200	$\sim$1 594	8–32

Stratified subsampling.

- **Children**: 4 age-bins on median admin age, equal n per bin; favor those with most admins / longest spans.
- **Items**: 4×4 stratification on (lexical class $\times$ $\log p_j$ quartile). 200 items kept.

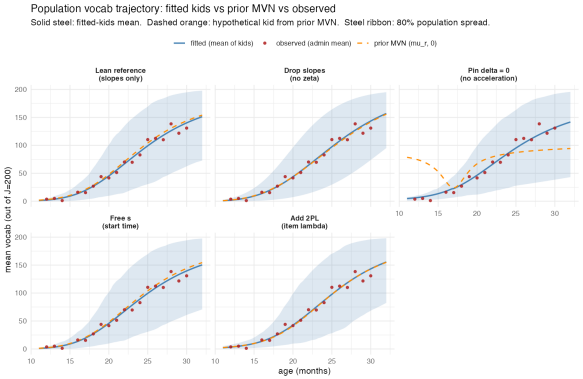
Stan. `log_irt_long.stan`; 4 chains $\times$ 1000 iter $\times$ 500 warmup; adapt_delta $= 0.95$; runs in 3–8 hours on Sherlock.

Results: lean baseline (English, slopes only)

Headline scalars (post. med. [95% Crl]):

Population		
σ_α	1.83	$[1.65, 2.04]$
σ_ξ	1.90	$[1.73, 2.11]$
σ_ζ	3.48	$[3.05, 3.96]$
$\rho_{\xi\zeta}$	0.40	$[0.27, 0.53]$
δ	9.39	$[8.58, 10.2]$
s	0.52	(pinned)
$\boldsymbol{\pi_\alpha}$	**0.913**	$\mathbf{[0.901, 0.926]}$

Efficiency variance dominates input variance: $\pi_\alpha \approx 0.91$.



Nested family: build the model one component at a time

We test each component's contribution by building a strictly nested spine. Every step *adds one named component*, and every step has a fit on disk so we can compare it to its predecessor by LOO ELPD.

stage	adds	interpretation	answers
M0	–	minimal IRT: per-child ξ_i, per-word ψ_j, no time, no frequency	identifiability anchor
M1	$+ \log \frac{t-s}{a_0}$	per-child accumulation grows with log-age (unit slope)	basic accumulator
M2	$+ \log p_j$	per-token frequency channel (unit coef)	**RQ2:** does freq matter?
M3	$+ \delta$	population-level age acceleration	**RQ1:** does growth accelerate?
M4	$+ \zeta_i$	per-child slope deviation around δ	between-child rate variance
M5	$+ \beta_c \cdot \log p_j$	class-specific frequency coefficient	class $\times$ freq interaction
M6	$+ \lambda_j$	per-word discrimination (2PL)	item-level sharpness

The next 7 slides walk through each stage's linear predictor, with the newly-added term highlighted.

M0: minimal IRT (no time, no frequency)

$$\eta_{ijt} = \xi_i + \log H - \psi_j$$

ξ_i $\log H$ ψ_j

child intercept (level only; absorbed by $\log r_i$ and $\log \mu_i$)

waking hours per word log exposure (free with class hierarchy)

threshold (free with class hierarchy)

What this is. The simplest possible item-response model: a fixed-in-age "ability" per child, a fixed difficulty per word, no mechanism. Vocabulary curves are flat in age; differences across children all live in ξ_i.

Why M0 first. Establishes the identifiability anchor for ξ_i (free per child) and ψ_j (free per word with class prior). Subsequent stages add structural channels on top of these.

$\mathbf{M_1 = M_0 + \text{time}}$

$$\eta_{ijt} = \xi_i + \log H + \mathbf{log}\frac{\mathbf{t-s}}{\mathbf{a_0}} - \psi_j$$

Time term: the child's accumulation grows with the log of (age − start time), at a fixed unit slope.

$$\log \frac{t-s}{a_0}$$

one log-token of evidence per log-age unit (no acceleration); a_0 is the per-dataset reference age (median admin); s is the global start time, pinned near 0 in the lean baseline.

What's pinned. $\delta = 0$ (no acceleration), $\beta_c = 0$ (no frequency channel), $\zeta_i = 0$, $\lambda_j = 1$. Equivalent to McMurray's (2007) deterministic accumulator without the frequency proxy or any per-child age dynamics beyond a constant log-age slope.

$\mathbf{M_2 = M_1 + \text{frequency}}$

$$\eta_{ijt} = \xi_i + \log H + \mathbf{\log p_j} + \log \frac{t-s}{a_0} - \psi_j$$

Frequency term: per-token corpus frequency of word j in CHILDES, with unit coefficient ("one log-token of evidence per log-token of corpus probability").

$$\log p_j$$

Coefficient is exactly 1 (the "unit-accumulator" assumption). M_2 vs. M_1 is the on-spine RQ2 test: does the data prefer a freq-aware model over per-word ψ_j alone?

What's still pinned. $\delta = 0$, $\zeta_i = 0$, $\beta_c = 1$ (unit), $\lambda_j = 1$.

$M_3 = M_2 + \delta$ (acceleration)

$$\eta_{ijt} = \xi_i + \log H + \log p_j + (1+\boldsymbol{\delta})\log\frac{t-s}{a_0} - \psi_j$$

Acceleration: the population log-age slope becomes $1+\delta$ instead of 1. With $\delta > 0$, evidence accumulates super-linearly in log-age $\Leftrightarrow$ token weight grows with development.

$$\delta$$

single global acceleration parameter; prior $\mathcal{N}(0, 0.5^2)$. M_3 vs. M_2 is the on-spine RQ1 test: must the slope deviate from 1 to fit observed growth-curve shape?

What's still pinned. $\zeta_i = 0$, $\beta_c = 1$, $\lambda_j = 1$.

$\mathbf{M_4 = M_3 + \zeta_i}$ (per-child slopes)

$$\eta_{ijt} = \xi_i + \log H + \log p_j + (1+\delta+\zeta_i)\log\frac{t-s}{a_0} - \psi_j$$

Per-child slope deviation: on top of the population acceleration, each child has their own ζ_i — a deviation of the log-age slope around δ.

$$\zeta_i$$

jointly drawn with ξ_i: $(\xi_i, \zeta_i) \sim \mathrm{MVN}(0, \Sigma)$ with LKJ prior on the correlation. Sum-to-zero centered, so δ stays separately identified from $\mathrm{mean}(\zeta_i)$.

What's still pinned. $\beta_c = 1$, $\lambda_j = 1$. Adding ζ_i requires *longitudinal* data to identify; the cross-sectional version of this fit is unidentified.

$\mathbf{M_5 = M_4 + \beta_c}$ (class-specific freq)

$$\eta_{ijt} = \xi_i + \log H + \boldsymbol{\beta_{c(j)}} \log p_j + (1+\delta+\zeta_i) \log \frac{t-s}{a_0} - \psi_j$$

Class-specific frequency coefficient: the unit weight on $\log p_j$ becomes a per-class free parameter β_c, one for each lexical class (nouns, predicates, function words, other).

$$\beta_c$$

prior $\mathcal{N}(1, 0.5^2)$ per class; $\beta_c = 1$ recovers M_4; $\beta_c < 1$ means the frequency channel contributes less than the unit-accumulator assumption for that class.

What's still pinned. $\lambda_j = 1$.

$\mathbf{M_6 = M_5 + \lambda_j}$ (2PL discrimination)

$$\eta_{ijt} = \boldsymbol{\lambda_j} \left[\xi_i + \log H + \beta_{c(j)} \log p_j + (1+\delta+\zeta_i) \log \frac{t-s}{a_0} - \psi_j \right]$$

Per-word discrimination: the entire ability − difficulty gap is multiplied by a word-specific λ_j. High λ_j = sharp transition near ψ_j (concrete nouns); low λ_j = gradual transition (function words, abstracts).

$$\lambda_j$$

hierarchical: $\log \lambda_j \sim \mathcal{N}(0, \sigma_\lambda^2)$ *with* $\sigma_\lambda \sim \mathrm{HN}(0,1)$. *Recovers the standard 2PL IRT extension.*

Full model. M_6 has the complete linear predictor as introduced on slide 5; everything to its left is now structurally free (or unit-anchored where the data prefer that).

Spine LOO comparison: English, $N = 200$ children

Step-wise ELPD differences (later − earlier; $\sigma_r = 0.534$ pinned):

step	ΔELPD	SE	z	verdict
M_1 vs M_0	+8 033	36	222.6	+ time: **matters**
M_2 vs M_1	**+2.2**	**1.85**	**1.2**	**+ frequency: null (RQ2)**
M_3 vs M_2	**+23 289**	**178**	**131.0**	**+ δ: matters, decisively (RQ1)**
M_4 vs M_3	+2 212	69	31.9	+ per-child slopes
M_5 vs M_4	+1.5	1.25	1.2	+ class β_c: *null*
M_6 vs M_5	+1 079	52	20.8	+ 2PL: marginal but real

Three closures. RQ1 (does δ matter?): closed at $z = 131$ – the load-bearing addition. **RQ2** (does frequency matter on-spine?): closed at $z = 1.2$ – frequency contributes nothing structural over and above per-word ψ_j. The M4-level off-spine variant agrees ($z = 0.5$). **Class** β_c: also null ($z = 1.2$). The per-class posteriors $(0.55, 0.45, 0.36, 0.16)$ trade cleanly with ψ_j – real-looking within fit, zero out-of-sample value.

Best by LOO: M_6. **Recommended adopt-and-augment scaffold:** M_4 (the slopes level), and the no-frequency variant for M4 best on input-uptake fits.

Spine LOO comparison: Norwegian, $N = 200$ children

Step-wise ELPD differences (some steps merged; M_3–M_5 Norwegian fits are pre-LOO rstan and lack `log_lik`):

step	ΔELPD	SE	z	verdict
M_1 vs M_0	+59 136	233	253.6	+ time alone: **huge**
M_2 vs M_1	**−46.5**	**4.5**	**−10.3**	**+ frequency: hurts (!)**
M_6 vs M_2	+4 215	84	50.1	+ δ + slopes + β_c + 2PL (lumped)
no_freq vs M_6	−1 423	59	−24.3	dropping freq from M_6: *hurts*

Norwegian-specific finding: degenerate M_1/M_2. With δ pinned at 0, the model in Norwegian collapses σ_α to ~ 0.03 and pushes s past its prior to ~ 2.0. Production variance gets absorbed into σ_r plus a global s-shift; $\pi_\alpha \to 0.003$ in M_1/M_2. Once δ is freed (in M_3 onward), σ_α recovers to ~ 1.57 and $\pi_\alpha \to 0.90$. The −46.5 ELPD step at M_2 vs M_1 is a comparison of two near-degenerate fits in this corner – read it as "adding unit-coef frequency in this regime makes a worse fit slightly worse," not as a substantive RQ2 finding on its own.

The cleaner Norwegian RQ2 test is no_freq vs M_6: with the full structure on top, dropping $\log p_j$ *hurts* fit by $\sim 1\,400$ ELPD ($z = -24$).

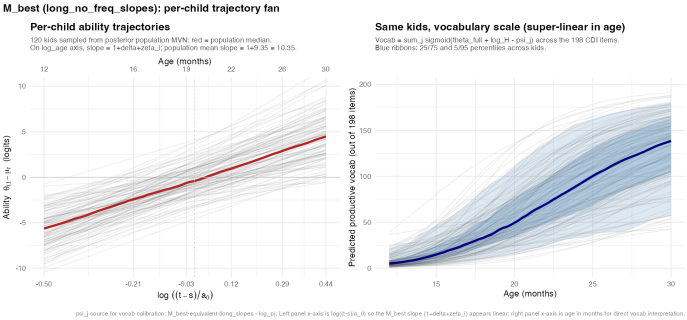
Cross-language reading: RQ1 universal, RQ2 culturally-specific

RQ test	English	Norwegian
RQ1: $+\delta$ on top of time+freq	$z = 131$	$z = 50$ (lumped)
RQ2 on-spine ($+\log p_j$ over time-only)	$z = 1.2$ *null*	$z = -10.3$ *neg*
RQ2 robustness (drop $\log p_j$ at top of spine)	$z = 0.5$ *null*	$z = -24.3$ *neg*

Reading. RQ1 (acceleration) is universal – both languages need $\delta \gg 0$ to fit the data, with σ_α and π_α both reaching plausible values once δ is freed.
RQ2 (frequency) is asymmetric across languages. In English, frequency is structurally redundant: the per-word ψ_j absorbs it. In Norwegian, dropping frequency at the top of the spine *hurts* fit by $\sim 1\,400$ ELPD – frequency is doing real work there.
Plausible mechanism. Norwegian Wordbank items match imperfectly to CHILDES Norwegian tokens (mixed Bokmål/Nynorsk orthography; many items fall back to a min-prob floor). At unit-β_c this noisy signal misleads M_2 (hence the -47 on-spine ELPD). Once class-specific β_c and per-word λ_j are free in M_6, the model can down-weight the noisy channel and freq becomes useful again. **The asymmetry is most likely about freq-data quality, not about a deep structural difference between the two languages.**

Headline: cross-linguistic, σ_α and $\pi_\alpha \approx 0.91$ replicate; δ varies (English ~ 9.4, Norwegian ~ 11.4); frequency's structural role is small everywhere and zero in English.

M_best: per-child trajectory fan



Reading.

- 120 kids drawn from posterior population MVN of M_best (`long_no_freq_slopes`). Each kid has both a baseline ξ_i and a growth-rate deviation ζ_i; kids fan out continuously.
- *Left*: $\theta_{it} - \mu_r$ on logit scale vs. $\log_{age}$ — trajectories visibly linear, slope $1 + \delta + \zeta_i$, mean ≈ 10.4.
- *Right*: same kids on linear age, vocab $\sum_j \sigma(\theta_{it} + \log H - \psi_j)$ over 198 CDI items; median tracks Wordbank ($\sim$150 at 30 mo).
- **Super-linearity** is visible *on the vocab axis*; on the logit axis it manifests as a steep slope ($\sim$10) on $\log_{age}$.

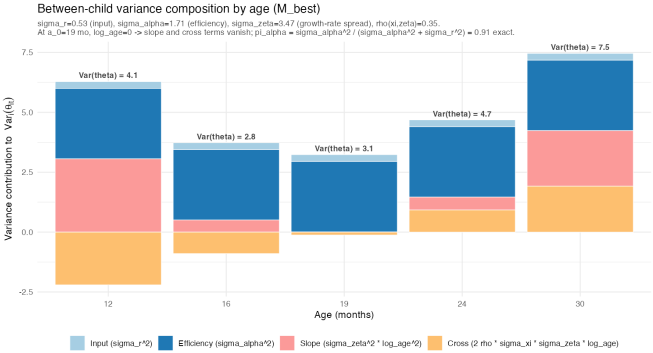
M_best: parameter posteriors

Param	Posterior	Units	What it does
μ_r	7.34 (pinned)	log tok/hr	Geometric mean input rate ($\sim$ 1 500 tok/hr).
σ_r	0.534 (pinned)	log scale	Between-child input variation (Sperry/HR/WF).
σ_α	1.71 [1.55, 1.91]	logits	Between-child efficiency variation.
σ_ξ	1.80 [1.64, 1.98]	logits	Total intercept variance: $\sigma_\xi^2 = \sigma_\alpha^2 + \sigma_r^2$.
$\boldsymbol{\pi_\alpha}$	**0.91 [0.89, 0.93]**	proportion	**Share of intercept variance from efficiency.**
δ	9.35 [9.21, 9.51]	dimensionless	Population age-acceleration: ability $\propto T^{1+\delta}$.
σ_ζ	3.47 [3.14, 3.84]	logits/log_age	Between-child variation in growth rate.
$\rho(\xi, \zeta)$	$+0.35$ $[+0.22, +0.47]$	correlation	Higher-baseline kids also have steeper growth.
s	0.53 (pinned ~ 0)	months	Global accumulation start age.

Three load-bearing reads.

- $\pi_\alpha = 0.91$: efficiency dominates input; $\sim$9% of intercept variance is CDS quantity.
- $\delta \approx 9.4$: vocabulary scales as T^{10} on cumulative tokens. McMurray's pure unit accumulator is $\delta = 0$.
- $\sigma_\zeta > \sigma_\alpha$: kids vary more in *growth rate* than in *baseline quality*.

M_best: where between-child variance comes from, by age



Decomposition.

$$\begin{aligned}\mathrm{Var}_i(\theta_{it}) &= \sigma_r^2 + \sigma_\alpha^2 \\ &\quad + \sigma_\zeta^2 \log_{age}^2 \\ &\quad + 2\rho\,\sigma_\xi\,\sigma_\zeta\,\log_{age}\end{aligned}$$

$$\log_{age} = \log\left(\frac{t-s}{a_0}\right).$$

Reads.

- At $a_0 = 19$ mo, $\log_{age} = 0$: only input + efficiency contribute. π_α exact here.
- Off-pivot: growth-rate variance grows quadratically; by 30 mo it dominates intercept components.
- Cross-term sign tracks $\log_{age}$: negative younger, positive older.
- π_α is an *intercept-share*, not a total-variance share — exact at a_0, qualified elsewhere.

Why we want input-uptake datasets

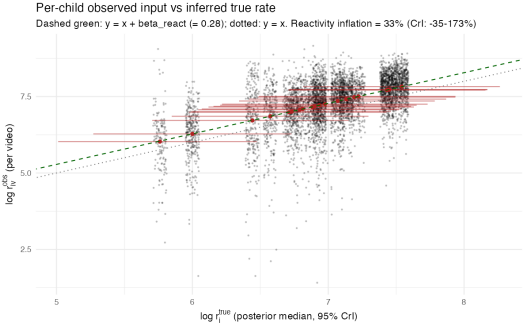
The longitudinal fit **infers** σ_r from external Sperry-style priors. To *measure* between-child input variance directly, we need datasets with per-child observed input rates.

	BabyView (Long et al. 2025)	**SEEDLingS** (Bergelson; Egan-Dailey 2025)
Recording channel	head-mounted body camera + transcript	passive LENA all-day audio
Observer effect	**yes** (β_{react} free)	**no** (β_{react} pinned at 0)
N kids (post-link)	20	44
CDI form	WG $\cup$ WS, longitudinal	WG, 6–18 mo
Per-recording rate	$\log r_v^{\text{obs}}$ from FEM/MAL tokens	$\log r_v^{\text{obs}}$ from LENA AWC

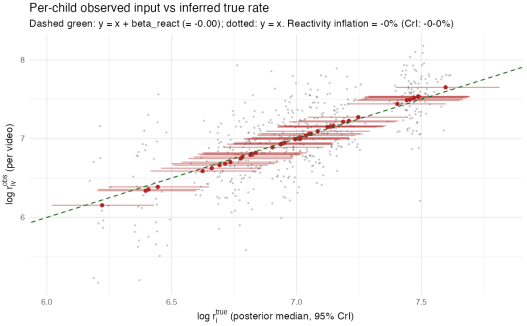
Modeling. Same Stan model (`log_irt_io.stan`); β_{react} on/off via dataset-level prior.

IO results: π_α replicates across samples

BabyView io_slopes (N=20)



SEEDLingS io_slopes (N=44)



dataset	N kids	π_α	δ	σ_α
English (long_slopes)	200	0.91 [0.90, 0.93]	9.39	1.83
BabyView (io_slopes)	20	0.86 [0.73, 0.94]	6.40	1.20
SEEDLingS (io_slopes)	44	**0.99 [0.99, 1.00]**	8.04	2.59

Three samples, two input-observation methods, one unobserved-input baseline: efficiency variance dominates.

Peekbank experiment: LWL as a window onto $\log \alpha$

Sample. 62 Adams & Marchman (2018) children with both LWL RT/accuracy from Peekbank (247 admins, 13–20 mo) and item-level CDIs at 22 + 25 mo from the Stanford TotLot 3 cohort.

Joint model.

$$\log \mathrm{rt}_{i,t} \sim \mathcal{N}\big(\mu_{rt} + \mu_{\mathrm{rtslope}}\,\ell_t \;-\; \boldsymbol{\gamma_{\mathrm{rt}}}\,\log \alpha_i \;+\; \ell_t\,\mathrm{rtslope}_i,\ \sigma_{\mathrm{lwl}}\big)$$

where $\ell_t = \log(t/a_0)$. $(\log \alpha, \zeta, \mathrm{rtslope}) \sim \mathrm{MVN}$ with LKJ prior on the correlation matrix.

Result.

- $\boldsymbol{\gamma_{\mathrm{rt}} = 0.073}$ $[0.032, 0.117]$ – bounded away from zero. Each unit of $\log \alpha$ corresponds to ~12% lower mean RT.
- $\mu_{\mathrm{rtslope}} = -0.73$: RT *decreases* with age, expected.
- $\sigma_{\mathrm{rtslope}} = 0.74$: real per-child variation in RT-improvement rate.

Comparison to prior accumulator models

Model	Per-child	Per-word	Acceleration	Bayesian	Real units
McMurray 2007	–	$g(\tau)$	via g'	–	–
Mitchell & McMurray 2009	–	$g(\tau)$ + leverage	via g'	–	–
Hidaka 2013	–	(N, δ, D)	via D	–	–
Mollica & Piantadosi 2017	–	(k_w, λ_w)	–	✓	partial
Kachergis et al. 2021	θ_i	d_j, class	–	–	–
Ours (M5)	$\xi_i, \zeta_i, \pi_\alpha$	$\psi_j, \lambda_j, \beta_c$	δ free	✓	✓

What's new in our nested family.

- McMurray 2007 / Mitchell & McMurray 2009: deterministic, no per-child variation. Their $g'(\tau) > 0$ requirement maps onto our **M2 vs. M1** test (does $\delta > 0$?).
- Hidaka 2013: closed-form Gamma / Weibull AoA distributions, population-averaged. Our **M1, M2, M3** occupy the same family (cumulative, rate-change, both) at the per-(child, word, age) Bernoulli level.
- Mollica & Piantadosi 2017: (k_w, λ_w) Gamma waiting time; large-k approximated by our logistic. Their joint (k, λ) ridge is our (s, δ) ridge.
- Kachergis et al. 2021: standardized logit IRT. Our model adds external σ_r pin (RQ3), structural δ (RQ1), real token units, and observation-channel architecture (uptake / processing).

Conclusions

- $\pi_\alpha \approx 0.86$–0.99 *across three samples*: between-child vocabulary variance is driven primarily by efficiency, not input rate.
- $\delta \approx 6$–9 *across all variants*: vocabulary growth genuinely accelerates with age; pinning $\delta = 0$ breaks population fit.
- LWL processing speed informs $\log \alpha$ in the expected direction; coupling $\gamma_{rt} > 0$ is bounded away from zero.

Code, data, fits, and full explainer: `github.com/langcog/standard-model-2`